Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	27 October 2026
Team ID	LMP-EV-2026
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language and logic used to present the website and perform BI analysis.	HTML5 and CSS3 for the static website; Tableau native aggregations and calculated fields for analysis. No separate Python or SQL application is evidenced.	HTML/CSS build the public landing page. Tableau performs measure aggregation, visualization and dashboard interactions without a separate analytics backend.
2.	Database / Data Warehouse	Structured source data used for analysis and reporting.	ev_market_train.csv (connected to Tableau); ev_market_test.csv (supplied as a resource but not connected); Microsoft Excel for dataset inspection. No database server or cloud warehouse is implemented.	CSV is portable and can be connected directly to Tableau. Excel is used to inspect records and fields. CSV/Excel are file sources, not a database service.
3.	Data Ingestion	Loads the connected training CSV into Tableau Desktop for field-level analysis.	Tableau Desktop text-file / CSV data connection.	Connects the EV market CSV to the workbook so the fields can be used in worksheets. Source replacement/reconnection is manual.
4.	Data Transformation and Business Logic	Configures fields, dimensions, measures, aggregations and analytical calculations.	CSV data preparation: remove null values, remove duplicate records and correct data types; Tableau Desktop field configuration, native SUM, AVG, MEDIAN, COUNTD and percentage-of-total calculations where configured.	Prepares the training data for consistent measures and dimensions before Tableau analysis. The test file is not used by the published workbook.
5.	Data Visualization / Analytics	Creates interactive charts, dashboards, summary KPIs and story views.	Tableau Desktop and Tableau Public (nine worksheets and dashboard; Story Point 1 completed).	Tableau Desktop is used to build the nine worksheets and dashboard. Tableau Public hosts the published workbook. Only Story Point 1, Global EV Market & OEM Overview, is complete; Story Points 2-4 are planned.

6.	Front-End / User Interface	Provides public entry to the project and access to interactive visualizations.	HTML5, CSS3 and GitHub Pages; Tableau Public embedded through an iframe.	The static website provides project navigation and embeds the published dashboard. Flask and Streamlit are not part of the evidenced deployed solution.

Architecture summary

ev_market_train.csv -> data cleaning (null/duplicate removal and data-type correction) -> Tableau Desktop data connection -> field configuration and aggregations -> 9 worksheets and dashboard -> completed Story Point 1 -> Tableau Public -> iframe embedded on the GitHub Pages website.

Website: https://en25cs3011105-spec.github.io/Electric-Vehicle-Market-Performance-and-Insights/

Tableau view: https://public.tableau.com/views/ElectricVechileMarketTechnologyAnalysis/ElectricVehicleMarketPerformanceInsights

Stack accuracy note: Only technologies evidenced by the project description and screenshots are listed as implemented. Python, SQL, Flask, Streamlit, Snowflake, MySQL and a dedicated database have not been added as implemented components.

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Confirmed source and scope: ev_market_train.csv is the connected source. ev_market_test.csv was supplied but not connected. The price_usd field is denominated in USD ($). Only Story Point 1 has been completed.